

Review

Health state utility values for patients with type 2 diabetes in China: A systematic review and meta-analysis

Xu Jin^{a,2}, Jiaxuan Shi^{a,2}, Yujia Hu^{b,2}, Haofei Li^a, Runhong Li^a, Yiyin Cao^a, Zitong Li^a,
Xiuzhi Cai^{c,*}, Weidong Huang^{a,1,*}

^a School of Health Management, Harbin Medical University, Harbin 150081, China

^b School of Nursing, Harbin Medical University, Harbin 150081, China

^c Harbin Municipal Center for Disease Control and Prevention, Harbin 150056, China

ARTICLE INFO

Keywords:

Health state utility values

Type 2 Diabetes Mellitus

Systematic review

ABSTRACT

This study performs a systematic review and *meta-analysis* to synthesize evidence on Health state utility values (HSUVs) for patients with type 2 diabetes mellitus (T2DM) in mainland China, aiming to inform cost-utility analyses and chronic disease management strategies. A systematic search of five major databases was conducted up to April 2025, with the review reported in accordance with PRISMA guidelines, aiming to identify studies reporting HSUVs for patients with T2DM and related complications and comorbidities in China. Relevant HSUVs were extracted and the random-effects model was used for *meta-analysis*, and the quality of included studies was evaluated using a four-item tool adapted for assessing HSUVs studies. In total, 35 studies were included in the final analysis. The pooled HSUV for patients with T2DM was estimated at 0.867 (95 % CI: 0.841–0.894). The presence of diabetes-related complications and comorbidities markedly reduced utility values, with HSUVs ranging from 0.673 to 0.828. Among the various measurement tools, the EQ-5D-3L was the most commonly used instrument, and considerable variation in utility estimates was observed across different valuation instruments, ranging from a low of 0.725 obtained using the SF-6D to a high of 0.968 using the 15D. This study provides a comprehensive synthesis of HSUVs for patients with T2DM in China, offering policymakers a nuanced understanding of HSUV heterogeneity to support the development of more targeted and effective healthcare interventions.

1. Introduction

Diabetes mellitus (DM) comprises a group of metabolic disorders characterized by chronic hyperglycemia [1]. T2DM as the most common form, arises primarily from insulin resistance and relative insulin secretion deficiency [2]. According to the 2021 World Diabetes Atlas, China has the largest diabetic population worldwide, with 141 million adults aged 20 and above affected by the disease. Notably, 51.7 % of these cases remain undiagnosed [3]. The T2DM population in China is marked by high prevalence, substantial underdiagnosis, and a trend toward earlier onset. Furthermore, T2DM is often accompanied by multiple complications [4], including both macrovascular and microvascular events—with stroke, chronic kidney disease, and retinopathy being the most common (incidence rates of 15.6 %, 11.5 %, and 9.3 %,

respectively) [5]. Since most complications are asymptomatic in early stages and screening rates remain low [6], many patients experience significant deterioration in health status and quality of life, presenting a serious public health challenge.

The economic burden imposed by T2DM and its complications in China places considerable strain on the healthcare system. Annual medical spending per T2DM patient rose from US\$1,292.72 in 2014 to US\$2,092.87 in 2019 [7], representing an annual growth rate of 12.38 % that surpassed the GDP growth rate during the same period [8]. Hospitalization costs for acute complications now exceed ¥6000 per admission, while annual costs for patients with macrovascular and microvascular complications reach ¥182,000 and ¥218,000, respectively [5]. These trends underscore the urgent need for improved prevention and management of T2DM. Moreover, employing economic

* Corresponding author.

E-mail addresses: caixiuzhi2008@163.com (X. Cai), huangweidong@hrbmu.edu.cn (W. Huang).

¹ Weidong Huang is the leading corresponding author.

² Xu Jin, Jiaxuan Shi, Yujia Hu have contributed equally to this work and share first authorship.

evaluation tools to assess the cost-effectiveness of interventions is essential for informing evidence-based health policy development.

Cost-utility analysis (CUA) is a widely used method in health economics [9]. Unlike cost-effectiveness analysis (CEA), CUA incorporates HSUVs, which allow comparisons across diverse health interventions [10]. In CUA, HSUVs serve as a key parameter reflecting an individual's health-related quality of life (HRQoL) across different health states, typically on a scale from 0 (death) to 1 (full health) [11]. For patients with T2DM, obtaining accurate HSUVs not only helps to evaluate the economic feasibility of various intervention measures (such as drug treatment, lifestyle intervention, complication prevention, etc.), but also provides a scientific basis for formulating more economical T2DM management policies.

Therefore, obtaining HSUVs that are representative of Chinese patients with T2DM is of critical importance. However, substantial variations exist across studies due to the interplay of disease-related factors (e.g., complications, disease duration), patient characteristics (e.g., age, comorbidities, socioeconomic status), and the choice of measurement instruments. Current evidence in this field is predominantly derived from Western cultural settings and healthcare systems, raising concerns about cross-cultural validity when generalized to Chinese populations [12–15]. Chinese T2DM patients often exhibit distinct symptom perceptions, treatment preferences, and quality-of-life values compared to Western groups. The direct use of utility data from other countries may therefore compromise the accuracy and relevance of health economic evaluations conducted in China. Furthermore, existing studies on HSUVs among Chinese T2DM patients are fragmented, lacking comprehensive synthesis and updated evidence, which limits their ability to accurately represent the current health status of this population [16–19].

To address these gaps in the literature, this study will employ systematic review and meta-analysis methods to synthesize and update the existing empirical evidence on HSUVs among patients with T2DM in mainland China. Specifically, the objectives are to estimate the range of HSUVs systematically and assess how population heterogeneity impacts these values. The findings are expected to provide robust evidence that will support economic evaluations and inform health policy development in China.

2. Methods

2.1. Search strategy and study selection

The search for studies adhering to our inclusion criteria was conducted in two distinct phases. Initially, We conducted a search of Web of Science, PubMed, Wanfang Data, the China Science and Technology Journal Database, and China National Knowledge Infrastructure (CNKI) from their inception until April 4, 2025. In the second stage, we performed a supplementary search to ensure the comprehensiveness of our literature collection. This process initially identified 33 eligible studies, with two additional studies retrieved from other sources, resulting in a final total of 35 included studies. These two supplementary studies were incorporated based on the systematic review by Wang et al. (2024) [12], which comprehensively summarized research on HSUVs for T2DM and its complications up to 2024. Given its broad and up-to-date coverage, the reference list of Wang et al. was used to verify and enhance the completeness of our study inclusion. The detailed search strategy can be found in the Supplementary file (Table S1-3).

This review was conducted to identify previously published studies reporting HSUVs for T2DM and its complications, and was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) [20]. To guarantee transparency and reproducibility, two independent reviewers carried out each stage. The initial screening involved a separate assessment of titles and abstracts to exclude studies that did not satisfy the inclusion criteria. For abstracts with insufficient information, full texts were retrieved for a complete assessment. Any discrepancies between the two reviewers were settled

through discussion or by consulting a third researcher. The protocol was PROSPERO-registered (CRD420251152726).

2.2. Inclusion and exclusion criteria

Eligibility for inclusion required studies to (1) Studies whose subjects were Chinese patients with T2DM, diagnosed according to the World Health Organization (WHO) criteria; (2) Studies that clearly reported HSUVs for the baseline condition or for complications; (3) Studies that used direct or indirect instruments to assess HSUVs. Studies were excluded for the following reasons: (1) Articles published in languages other than Chinese or English; (2) Duplicate publications or studies for which the full text was unavailable; (3) Studies that do not involve original human research, including reviews, reports, conference proceedings, and guidelines; (4) Studies that did not report any HSUVs; (5) Studies conducted on non-mainland Chinese populations, such as those in Hong Kong, Macau, or Taiwan.

2.3. Data extraction

For each study, we extracted and recorded the following information into a Microsoft Excel spreadsheet: (1) Basic characteristics: first author, publication year, study period, study region, study type, sample size, and study population (proportion of female participants); (2) HSUVs measurement instruments; and (3) HSUVs for T2DM patients and their complications/comorbidities.

2.4. Quality assessment

Considering that relying solely on study design to select a quality assessment tool may not fully reflect the quality of the literature [21], this study adapted a quality assessment tool developed by Yubo Wang et al. for use in this systematic review [12]. This tool [22] is based on 17 evaluation items proposed by Nerich et al. and incorporates recommendations from a systematic review [23] by Zoratti et al. From these, four key questions were selected that are particularly suitable for assessing the quality of HSUVs literature (Table S5). Existing research has shown that these four items are highly specific for identifying the internal validity of a study, which helps in systematically evaluating utility studies from two perspectives: methodological rigor and result applicability [23]. Each item's answer options include "Yes" and "No." The "Yes" option is further subdivided into "complete" and "partial" to reflect the completeness of the reported information. This tool addresses the shortcomings of existing quality assessment methods for HSUVs research, providing a more targeted basis for quality judgment in this study.

2.5. Statistical analysis

This study performed a meta-analysis using the meta and metafor packages in R, with the effect size (ES) and its 95 % confidence interval (CI) as the effect measures [24]. Inter-study heterogeneity was assessed using Cochran's Q test (with a significance level of $\alpha = 0.10$) and the I^2 statistic. A random-effects model was used for the pooled analysis when significant heterogeneity was present ($P < 0.10$ and $I^2 > 50\%$); otherwise, a fixed-effects model was adopted. When ten or more studies were included, a funnel plot was also generated and Egger's regression test was performed to evaluate the potential for publication bias.

3. Result

3.1. Study selection

A total of 11,142 relevant studies were retrieved. After an initial screening of titles and abstracts, 7418 were retained. The initial screening process excluded 7275 articles, including 606 reviews, 6668

articles without HSUVs, and one study focused on caregivers. A full-text review of the remaining articles led to the exclusion of another 110 for not containing HSUVs. This process yielded 33 included articles, with two additional studies identified from other sources, bringing the final total to 35. These two supplementary studies were incorporated based on the findings of the systematic review by Wang et al. (2024). The complete study selection process is detailed in the PRISMA flow diagram (Fig. S1).

3.2. Quality assessment

The quality assessment revealed several reporting gaps (Table S6). Specifically, nine studies did not explicitly justify their choice of HSUVs measurement tool, while three provided insufficient detail on the tool's application. Furthermore, two studies lacked a clear rationale for their sample population selection. In contrast, a comprehensive description of respondent characteristics was consistently provided across all included studies. In summary, while the existing research demonstrates strong reporting on patient demographics, there is a clear need for improvement in justifying the theoretical basis for the selection of measurement instruments.

3.3. Study characteristics

The main characteristics of the included studies are summarized in Table 1. These studies represent a broad geographical distribution across China, encompassing both nationwide multi-center trials and regionally focused investigations conducted in specific provinces and municipalities. Among the included literature, eight studies adopted a nationwide design [17,25–31], while thirteen were conducted within specific provincial-level populations. The distribution of these provincial studies is as follows: five in Shandong Province [32–36], two in Jiangsu Province [37,38], and five additional studies each carried out in Liaoning [39], Anhui [40], Henan [41], Xinjiang [42], and Hainan Province [43], respectively. One further multi-center study covered Shanghai, Jiangsu, Zhejiang, and Jiangxi Provinces [44]. Additionally, fourteen studies were situated in specific municipalities, distributed as follows: four in Suzhou Municipality [45–47], two multi-center urban investigations encompassing Beijing, Guangzhou, Nanjing, Chengdu, and Shenyang [48] as well as nine cities across China [49], and eight studies each located in a single municipality, namely Shanghai [50], Tianjin and Beijing (combined in one study) [51], Qingdao [52], Changchun [53], Nanjing [54], Ningbo [55], Shenzhen [56], and Chengdu [56]. The overall study design was predominantly cross-sectional, and the average age of the included patient populations was approximately 60 years.

In terms of measuring HSUVs, the studies included in this analysis employed both direct and indirect valuation methods. Among the direct valuation methods, the most commonly used is the Time Trade-Off (TTO) method, which guides participants to make trade-offs between life expectancy and health status, allowing for the direct elicitation of individual preferences for specific health states. Indirect valuation methods include commonly used HSUVs instruments such as the three-level EuroQol Five-Dimensional Questionnaire 3-Level (EQ-5D-3L), the five-level EuroQol Five-Dimensional Questionnaire 5-Level (EQ-5D-5L), the Short Form 6-Dimension version 1 (SF-6Dv1), the Short Form 6-Dimension version 1 (SF-6Dv2), and the 15D. Among the included studies, twenty utilized the EQ-5D-3L [17,25,27,28,30–34,37,39,40,44,47–49,54,56–58], fourteen employed the EQ-5D-5L [26,29,35,36,38,41–43,45,46,50,52,55,58], one study used the 15D [51], two applied the SF-6D [36,57], and one study adopted the TTO approach [53]. It is noteworthy that three studies applied two distinct preference-based measures concurrently: Chen-Wei Pan et al. (2015) [58] used both EQ-5D-5L and EQ-5D-3L; Huimin Du et al. (2023) [36] employed both EQ-5D-5L and SF-6Dv2; and Kun Xie et al. (2014) [57] used both EQ-5D-3L and SF-6Dv1. Regarding the value sets used for scoring health utilities, six studies applied the UK value set [26,27,39,40,49,57], one used the Finnish value set [51], and three

adopted the Japanese value set [29,30,44]. The majority of studies ($n = 22$) employed the Chinese value set [17,28,31–38,41–43,45–48,50,54–56,58], while three studies did not explicitly report the source of their value set [25,52,53]. Among these instruments, the EQ-5D is the most frequently used, with the majority of studies adopting the Chinese value set. The reported HSUVs range from 0.562 to 0.977. Table 2 shows the HSUVs for T2DM patients with different comorbidities and complications, with values ranging from 0.434 to 0.890. The most common comorbidity covered in the studies is ocular disease, while cardiovascular disease is the most frequently reported complication.

3.4. Health utility values

A total of 35 studies were included in this review; however, only 28 studies reported both means and standard deviations and were therefore included in the meta-analysis. The meta-analysis of these 28 studies revealed substantial heterogeneity ($Q = 12917.25$, $I^2 = 99.7\%$, $P < 0.001$), leading to the use of a random-effects model. The pooled HSUV for patients with T2DM was 0.867 (95 % CI: 0.841–0.894, $P < 0.001$), as presented in Fig. 1.

The meta-analysis of HSUVs for patients with T2DM and related complications/comorbidities revealed notable variations across conditions, age groups, patient sources, and measurement instruments (Table S7). Patients with stroke (0.673, 95 % CI: 0.570–0.776) and peripheral vascular disease (0.684, 95 % CI: 0.546–0.822) exhibited the lowest utility values, followed by those with diabetic foot (0.712, 95 % CI: 0.621–0.804) and nephropathy (0.750, 95 % CI: 0.701–0.798). In contrast, utility values were relatively higher among patients with retinopathy (0.851, 95 % CI: 0.802–0.899), hypertension (0.811, 95 % CI: 0.758–0.865), and hyperlipidemia (0.828, 95 % CI: 0.787–0.869).

Subgroup analyses highlighted the influence of patient characteristics and measurement instruments (Table S8). Age-stratified results showed a slight decline in utility values with increasing age: <55 years, 0.877 (95 % CI: 0.833–0.921); 56–60 years, 0.872 (95 % CI: 0.832–0.912); >60 years, 0.865 (95 % CI: 0.825–0.905). Regarding patient source, community populations reported the highest HSUVs at 0.926 (95 % CI: 0.877–0.974), whereas hospital populations had the lowest at 0.846 (95 % CI: 0.818–0.874). Patients sampled from both settings showed an HSUV of 0.783 (95 % CI: 0.628–0.937). Utility estimates also varied by measurement instrument. The 15D instrument yielded the highest value of 0.968 (95 % CI: 0.962–0.975), followed by EQ-5D-5L at 0.901 (95 % CI: 0.868–0.934), EQ-5D-3L at 0.852 (95 % CI: 0.814–0.889), TTO at 0.773 (95 % CI: 0.668–0.878), and SF-6D at 0.725 (95 % CI: 0.656–0.794).

In order to evaluate publication bias, a funnel plot was generated and Egger's regression test was performed. As shown in Fig. S24, the included studies were primarily clustered to the right of the pooled effect size, suggesting the potential for publication bias. Egger's test further confirmed this bias (intercept = 0.917, $Z = -5.87$, $P < 0.001$), suggesting that studies with significant or favorable findings were more likely to be published.

4. Discussion

As the first national systematic review and meta-analysis on HSUVs in patients with T2DM and its complications, this study provided comprehensive evidence and addressed a key research gap in the field. By synthesizing existing evidence, this study provided key parameters for localized CUA and clarified the significant effects of complication types, patient characteristics, and measurement instruments on HSUVs. The findings offered solid empirical support for developing more targeted and evidence-based strategies to improve the prevention, control, and management of T2DM in China.

Using a meta-analysis that integrated data from multiple studies conducted in mainland China, this study estimated a pooled HSUV of 0.867 (95 % CI: 0.841–0.894) for Chinese patients with T2DM. This

Table 1
Study characteristics.

Study ID	Study period	Study region	Study design	Age	Sample size	Patient resource	Sex, female n (%)	Instrument	HSUVs Mean (SD)/Median (Q1, Q3)	The value set
Chen, Z. et al. (2011) [25]	2007.1–2007.6	China	Cross-sectional study	61	486	Hospital	52.9 %	EQ-5D-3L	0.920(0.130)	–
Han, Y. et al. (2013) [49]	2008.12–2009.7	9 cities in China	Cross-sectional study	59.6	7,082	Hospital	48.3 %	EQ-5D-3L	0.870(0.210)	UK
Zhang, Y. et al. (2014) [51]	2009.4–2011.9	Qingdao Municipality	Cross-sectional study	59 (10.4) 58 (10.8) 60 (9.2) 59 (9.8)	132 399 114 106	–	65.2 % 50.4 % 60.5 % 78.3 %	15D	Newly diagnosed (urban): 0.975 (0.057) Newly diagnosed (rural): 0.972 (0.060) Previously diagnosed (urban): 0.964 (0.053) Previously diagnosed (rural): 0.960 (0.051) Baseline: 0.875 (0.179) After 24 weeks: 0.893 (0.160) 0.780(0.240) 0.760(0.120)	Finland
Yang, W. et al. (2014) [26]	2009.1–2010.6	China	Observational Study	54.9 (14.4)	8,578	Hospital	43.0 %	EQ-5D-5L	0.875 (0.179) 0.893 (0.160) 0.780(0.240) 0.760(0.120)	UK
Xie, K. et al. (2014) [57]	2011.7–2011.12	Tianjin and Beijing Municipality	Cross-sectional study	63.5	573	Hospital	55.3 %	EQ-5D-3L SF-6D-v1	0.900	UK
Ji, L. et al. (2015) [27]	2011.8–2012.3	China	Cross-sectional study	–	685	Hospital	–	EQ-5D-3L	0.900	UK
Pan, C-W. et al. (2015) [58]	2014.3–2014.6	Suzhou Municipality	Cross-sectional study	64.9 (9.1)	289	Hospital	69.4 %	EQ-5D-3L EQ-5D-5L	0.898(0.160) 0.876(0.140)	China mapping
Pan, C-W. et al. (2016) [46]	2014.3–2014.6	Suzhou Municipality	Cross-sectional study	–	289	Hospital	69.5 %	EQ-5D-5L	0.876(0.140)	China
Lu, Y. et al. (2017) [44]	2014	Shanghai, Jiangsu, Zhejiang and Jiangxi Provinces	Cross-sectional study	52.2 (13.4)	5,310	–	56.3 %	EQ-5D-3L	0.922(0.122)	Japan
Xu, L. et al. (2017) [32]	–	Shandong Province	Cross-sectional study	65.0 (9.14)	1,948	Community	66.5 %	EQ-5D-3L	0.851(0.186)	China
Li, Y. et al. (2017) [48]	–	Beijing, Guangzhou, Nanjing, Chengdu and Shenyang Municipality	Cross-sectional study	–	2,879	–	44.6 %	EQ-5D-3L	0.889	China
Huang, S. et al. (2017) [52]	2016.5–2016.11	Changchun Municipality	Cross-sectional study	51.6 (10.2)	260	Hospital	48.5 %	EQ-5D-5L	0.768(0.705, 0.804)	–
Pan, C-W. et al. (2018) [45]	–	Suzhou Municipality	Cross-sectional study	–	913	Community	55.9 %	EQ-5D-5L	0.983(0.067)	China
Jin, X. et al. (2018) [17]	2010.12–2011.10	China	Observational Study	61.2	969	Community	61.3 %	EQ-5D-3L	0.790	China
Wang, Z. et al.	–	Liaoning Province	Cross-sectional study	63.6 (11.8)	394	Hospital	52.0 %	EQ-5D-3L	0.750(0.330)	UK

(continued on next page)

Table 1 (continued)

Study ID	Study period	Study region	Study design	Age	Sample size	Patient resource	Sex, female n (%)	Instrument	HSUVs Mean (SD)/ Median (Q1, Q3)	The value set
(2018) [39] Xu, Q. et al.	2016.8–2016.10	Shandong Province	Cross-sectional study	—	1,948	Community	66.5 %	EQ-5D-3L	Rural: 0.840 (0.190) Urban: 0.880 (0.170)	China
(2019) [33] Li, M. et al.	—	Jiangsu Province	Cross-sectional study	—	181	Hospital and community	51.9 %	EQ-5D-3L	0.873(0.191)	China
(2019) [37] Wang, S. et al.	2015	Jiangsu Province	Cross-sectional study	67.7 (8.3)	880	Community	55.9 % 55.9 % 55.9 %	EQ-5D-5L	Without retinopathy: 0.983 (0.067) Unilateral retinopathy: 0.971 (0.082) Bilateral retinopathy: 0.970 (0.145)	China
(2019) [40] Wang, D. et al.	2016.09–2019.2	Anhui Province	Longitudinal Study	59.0 (2.9)	100	Hospital	Intervention (pre): 39.0 %	EQ-5D-3L	0.812(0.158)	UK
(2019) [40] Zhuang, Y. et al.	2014	Suzhou Municipality	Cross-sectional study	58.0 (10.1)	100	Hospital	Intervention (post): 36.0 %	EQ-5D-3L	0.848(0.175)	UK
(2020) [47] Zhang, Y. et al.	—	China	Cross-sectional study	68.6 (6.6)	220	Hospital	69.1 %	EQ-5D-3L	0.886(0.170)	China
(2020) [28] Tang, Z. et al.	—	China	Cross-sectional study	59.6 (12.6)	7,081	Hospital	49.1 %	EQ-5D-3L	0.870(0.150)	China
(2020) [50] Gu, S. et al.	2018.4–2018.5	Shanghai Municipality	Cross-sectional study	64.0 (9.7)	277	Hospital	42.6 %	EQ-5D-5L	0.904(0.132)	China
(2020) [34] Gu, S. et al.	2017.1–2017.5	Shandong Province	Cross-sectional study	—	802	Hospital	43.0 %	EQ-5D-3L	0.562	China
(2020) [53] Hao, S. et al.	2016	Nanjing Municipality	Cross-sectional study	59.9 (11.3)	80	Hospital	46.3 %	Iterative TTO	0.700(0.280)	—
				56.2 (13.0)	80		43.8 %	Open-ended TTO	0.870(0.180)	—
				55.9 (10.4)	33		36.4 %	Modified Open-ended TTO	0.740(0.240)	—
(2020) [29] Li, X. et al.	2015.3–2015.11	China	Cross-sectional study	67.6 (6.0)	479	Hospital	51.6 %	EQ-5D-5L	0.770(0.130)	Japan
(2020) [30] Li, X. et al.	2015.3–2015.5	12 Provinces and Municipalities in China	Cross-sectional study	54.7 (11.6)	69	Hospital	44.9 %	EQ-5D-3L	Lifestyle intervention group: 0.815 (0.076)	Japan
				56.7 (11.5)	515		44.9 %		Oral medication group: 0.800 (0.106)	
				59.2 (11.1)	479		45.5 %		Insulin therapy group: 0.772 (0.125)	
(2021) [31] Li, C. et al.	2015.10–2017.3	China	Observational Study	59.5 (11.9)	12,583	Hospital	47.4 %	EQ-5D-3L	Total population: 0.936 (0.120)	China
				59.7 (11.9)	6,827		47.4 %		Modeling population: 0.929 (0.127)	
				59.7 (11.9)	5,756		47.4 %		Validation population: 0.945 (0.111)	
Kang, N. et al.	2015.7–2017.9	Henan Province	Cross-sectional study	60.4 (9.55)	2,231	—	60.0 %	EQ-5D-5L	0.930(0.150)	China

(continued on next page)

Table 1 (continued)

Study ID	Study period	Study region	Study design	Age	Sample size	Patient resource	Sex, female n (%)	Instrument	HSUVs Mean (SD)/Median (Q1, Q3)	The value set
(2021) [41] Liu, Q. et al.	2021.6–2021.12	Xinjiang Province	Cross-sectional study	57.8 (11.2)	297	Hospital	37.1 %	EQ-5D-5L	0.893(0.876, 0.951)	China
(2022) [42] Carter, N. et al.	2020.11–2021.5	Ningbo Municipality	Cross-sectional study	—	406	Hospital	37.2 %	EQ-5D-3L	0.977(0.065)	China
(2022) [54] Zhao, M. et al.	2021.9–2021.12	Shenzhen Municipality	Cross-sectional study	56.5 (13.0)	634	Hospital and community	44.3 %	EQ-5D-5L	0.888(0.205)	China
(2023) [55] Liu, Y. et al.	2022.7–2022.9	Shandong Province	Cross-sectional study	61.3 (10.7)	397	Hospital and community	41.1 %	EQ-5D-5L	0.820(0.210)	China
(2023) [35] Lin, P. et al.	2021.9–2022 0.1	Hainan Province	Cross-sectional study	63.3 (11.7)	736	Hospital and community	47.7 %	EQ-5D-5L	0.907(0.782, 1.000)	China
(2023) [43] Du, H. et al.	2020.12—2021.7	Shandong Province	Cross-sectional study	57	1,084	Hospital	41.7 %	EQ-5D-5L SF-6Dv2	0.870(0.220) 0.690(0.160)	China
(2023) [36] Su, S. et al.	2022.3–2023.1	Chengdu Municipality	Cross-sectional study	—	433	Hospital and community	48.3 %	EQ-5D-3L	0.550(0.280)	China
(2023) [56]										

Note: “—” indicates not mentioned.

estimate is consistent with the findings of study that reported similar HSUVs for a T2DM subgroup within the Chinese diabetic population [19], thereby supporting the robustness of the result. Using a meta-analysis that integrated data from multiple studies conducted in mainland China, this study estimated a pooled HSUV of 0.867 (95 % CI: 0.841–0.894) for Chinese patients with T2DM. This estimate is consistent with the findings of study that reported similar HSUVs for a T2DM subgroup within the Chinese diabetic population [19], thereby supporting the robustness of the result. Compared with international data, the HSUVs of mainland Chinese patients were notably higher than those reported in several Western countries. For example, 0.790 (95 % CI: 0.780–0.800) in Canada and 0.720 (95 % CI: 0.640–0.790) in the UK and were closer to estimates from Japan (0.860, 95 % CI: 0.800–0.920) and pooled results across East and Southeast Asia (0.881, 95 % CI: 0.833–0.929)[14,59]. This regional variation may primarily reflect differences in demographic and clinical characteristics. An estimated 247.6 million people in East and Southeast Asia have diabetes, shouldering 53 % of the global burden. China accounts for the largest share of this population [14]. Evidence suggests that Chinese patients tend to develop T2DM at an earlier age and are diagnosed younger, resulting in fewer comorbidities and better functional status at onset [60]. Consequently, they are more likely to perceive their overall health positively, which in turn contributes to higher HSUVs. These findings highlight the need to develop and implement youth-oriented prevention, control, and management programs for T2DM. Addressing the earlier onset of disease requires shifting from a treatment-centered approach to a prevention-oriented model emphasizing early intervention and sustained management.

The presence of comorbidities and complications is well-established to diminish the quality of life and lower HSUVs in patients with T2DM, with reported utility decrements ranging from 0.045 to 0.180 [61]. Our findings align with this, showing a decrement range of 0.016 to 0.183 [62]. In our analysis, the most significant reductions in quality of life

were observed in patients with comorbid stroke and peripheral vascular disease. This is clinically plausible, as patients with diabetes have double the risk of stroke compared to the non-diabetic population, often at a younger age, leading to long-term disability and a marked decline in quality of life [63]. Moreover, peripheral vascular disease not only causes stenosis or occlusion of lower limb arteries, severely impairing mobility, but also significantly increases the risk of cardiovascular events and mortality, thus being considered one of the most severe complications. In contrast, one study reported that amputation resulted in the largest reduction in HSUVs [61], whereas another found that patients with comorbid kidney disease had the lowest HSUVs [64]. These discrepancies may be attributed to variations in study populations, diagnostic criteria, and assessment methodologies. In summary, early intervention and prevention of comorbidities and complications are essential for improving the quality of life among patients with T2DM. Clinical management should prioritize the prevention and early control of complications, particularly comorbid stroke and peripheral vascular disease. Comprehensive management of glycemic, blood pressure, and lipid control, in conjunction with rehabilitation therapy, can minimize the risk of functional impairment and disability, thereby enhancing patients' overall health status.

Our results indicate that older age is a significant determinant of HSUVs in T2DM patients, with those aged 60 and above reporting the lowest values 0.865 (95 % CI: 0.825–0.905). This trend can be attributed to a combination of factors; as patients age, they face declining basal metabolic function and an increased burden of chronic diseases. Concurrently, psychosocial factors such as social isolation and insufficient social support may contribute to a greater psychological burden, resulting in lower reported HSUVs [65]. This finding is consistent with a large-scale study of 4963 diabetic patients in Japan which, despite not reporting age-stratified HSUVs directly, demonstrated a declining trend in HSUVs with increasing age in their regression models [66]. Consequently, it is crucial to recognize the unique needs of elderly patients

Table 2
HSUVs of T2DM patients with different comorbidities/complications.

Study ID	Peripheral neuropathy	Peripheral vascular disease	Retinopathy	Nephropathy	Diabetic foot	Stroke	Hypertension	Hyperlipidemia	Cerebrovascular disease	Cardiovascular disease
Han, Y. et al. (2013) [49]	0.780 (0.240)	—	0.790 (0.250)	0.750 (0.280)	0.660 (0.340)	0.710 (0.330)	0.840 (0.230)	0.840 (0.220)	—	Heart disease: 0.790 (0.260)
Xie, K. et al. (2014) [57]	0.730 (0.230) 0.720 (0.100)	0.590 (0.350) 0.640 (0.150)	0.750 (0.250) 0.730 (0.120)	0.740 (0.240) 0.710 (0.110)	0.620 (0.290) 0.690 (0.110)	— —	0.750 (0.260) 0.740 (0.130)	0.770 (0.220) 0.790 (0.250)	0.730 (0.260) 0.730 (0.110)	0.690 (0.310) 0.720 (0.130)
Pan, C-W. et al. (2015) [58]	0.846 (0.210) 0.815 (0.170)	— —	0.870 (0.170) 0.843 (0.150)	— —	— —	— —	— —	— —	0.695 (0.350) 0.674 (0.290)	0.784 (0.250) 0.764 (0.210)
Pan, C-W. et al. (2016) [58]	0.815 (0.170)	—	0.843 (0.150)	—	—	—	—	—	—	—
Pan, C-W. et al. (2018) [45]	—	—	Unilateral: 0.971 (0.082) Bilateral: 0.970 (0.145)	—	—	—	—	—	—	—
Wang, S. et al. (2019) [38]	—	—	Unilateral: 0.971 (0.082) Bilateral: 0.970 (0.145)	—	—	—	—	—	—	—
Zhang, Y. et al. (2020) [28]	0.810 (0.170)	—	0.810 (0.180)	0.780 (0.200)	0.710 (0.230)	0.740 (0.240)	0.850 (0.170)	0.850 (0.160)	—	0.810 (0.190)
Gu, S. et al. (2020) [34]	—	—	0.714 (0.047)	0.693 (0.047)	0.395 (0.107)	0.472 (0.072)	—	—	—	Ischemic heart disease: 0.620 (0.045) Acute myocardial infarction: 0.434 (0.079) Congestive heart failure: 0.471 (0.077) 0.760 (0.140)
Li, X. et al. (2020) [29]	0.740 (0.160)	0.740 (0.170)	0.750 (0.140)	0.750 (0.140)	—	—	0.760 (0.140)	0.760 (0.140)	—	0.760 (0.140)
Li, C. et al. (2021) [31]	0.910 (0.144)	0.904 (0.158)	0.887 (0.141)	0.844 (0.185)	0.797 (0.219)	0.830 (0.198)	—	—	—	Heart failure: 0.750 (0.224) Myocardial infarction: 0.886 (0.154) Angina pectoris: 0.868 (0.177) Heart failure: 0.792 (0.083) Myocardial infarction: 0.887 (0.083) Coronary heart disease angina: 0.765 (0.067)
Zhao, M. et al. (2023) [55]	—	0.821 (0.055)	0.818 (0.041)	0.699 (0.110)	—	0.569 (0.067)	0.873 (0.014)	0.886 (0.012)	—	0.890 (0.200) 0.700 (0.160)
Du, H. et al. (2023) [36]	0.820 (0.250) 0.660 (0.160)	— —	0.870 (0.230) 0.700 (0.160)	0.880 (0.200) 0.690 (0.160)	0.890 (0.180) 0.700 (0.160)	— —	— —	— —	— —	0.890 (0.200) 0.700 (0.160)

Note: “—” indicates not mentioned.

when developing individualized management and health intervention strategies. Providing targeted support can help mitigate the decline in their quality of life and improve overall health outcomes.

HSUV estimates are significantly influenced by the measurement instrument employed. As a direct measurement method, Time Trade-Off (TTO) tended to yield lower HSUVs. This may be because the method forces respondents to make an explicit trade-off between length and quality of life, potentially amplifying the perceived impact of health decrements [67,68]. Among the indirect instruments, the 15D

instrument produced the highest HSUVs, followed by the EQ-5D-5L and EQ-5D-3L, while the SF-6D yielded the lowest. The choice of value set is also a critical determinant. For example, a comparative study showed that when using the Hungarian value set, the mean HSUV for the EQ-5D-5L was 0.91, whereas the mean for the 15D was only 0.81 with the Norwegian value set [69]. The lower HSUVs derived from the SF-6D may be related to its less pronounced ceiling effect compared to other instruments [70–73]. Interestingly, although the EQ-5D-3L is theoretically more prone to ceiling effects (which can overestimate health states), our

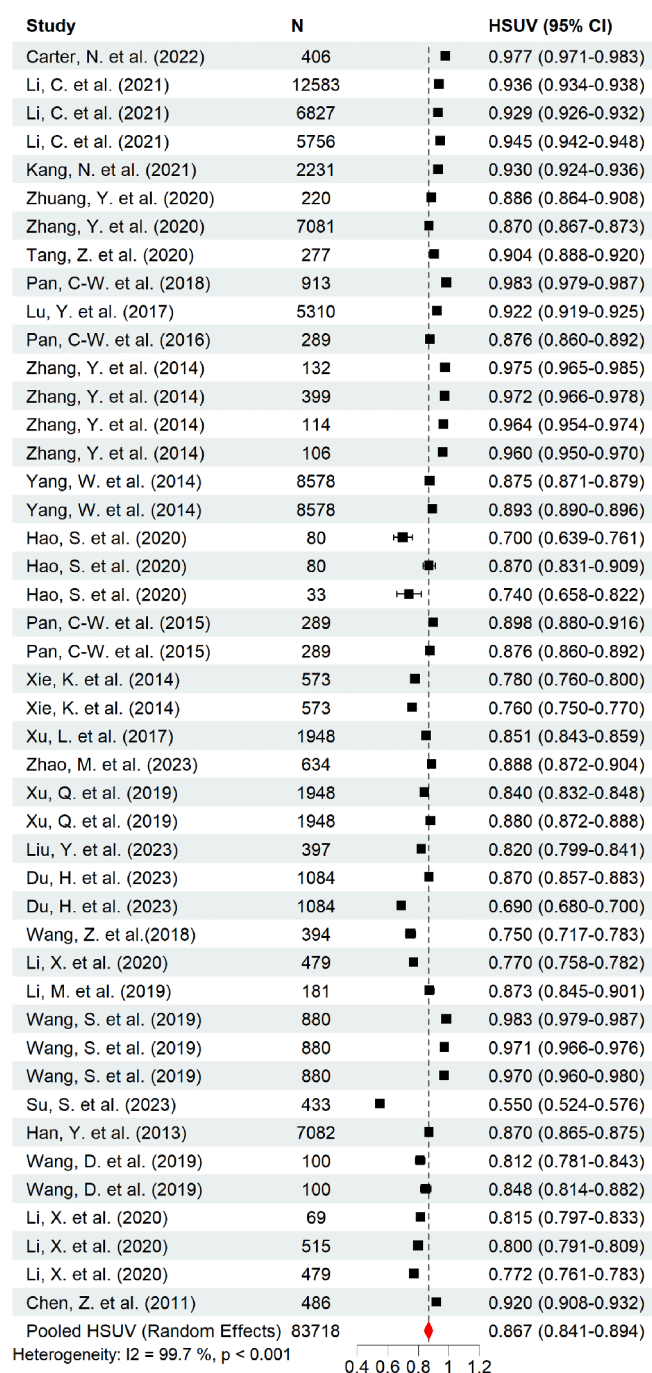


Fig. 1. Forest plot of HSUVs of T2DM.

meta-analysis found that the EQ-5D-5L produced higher utility values [74–78]. This counterintuitive finding might be a temporal artifact, as the EQ-5D-5L has been used more frequently in recent years, a period during which advances in medical care and disease management may have improved the overall health of T2DM patients [12]. Furthermore, HSUV estimates vary significantly depending on the national value set used; studies have shown that the Chinese and US value sets produce higher HSUVs than the UK and Japanese tariffs, reflecting different societal preferences for health states [79]. This underscores the importance of using country-specific value sets in health economic evaluations to ensure local relevance and accuracy [80]. In summary, differences in the dimensions, descriptive levels, and valuation systems of these instruments all contribute to variability in HSUV estimates and must be carefully considered during instrument selection.

Despite these strengths, several limitations should be acknowledged. First, our data search was restricted to published literature in English and Chinese. The exclusion of unpublished studies or grey literature may have introduced publication bias, potentially affecting the overall HSUVs estimate. Second, although we employed a simplified quality assessment tool tailored for HSUVs studies to enhance the focus on internal validity, this tool does not cover all potential sources of bias, such as selection and measurement bias, thus limiting a fully comprehensive quality evaluation. Finally, significant heterogeneity in HSUVs estimates persists due to the combined influence of different measurement instruments and value sets. While our subgroup analysis addressed the choice of instrument, further research with more granular data is needed to fully disentangle the complex effects of these factors.

5. Conclusion

This study systematically evaluated the HSUVs of patients with T2DM and its complications in China, providing comprehensive and locally relevant evidence for this field. Based on these findings, future clinical practice and public health policies should prioritize an integrated approach that combines medical treatment with prevention, emphasizing early screening and control of complications as well as individualized management for specific populations such as older adults, to improve the overall quality of life of patients with T2DM in China. In addition, future health economics research should prioritize the use of validated measurement instruments and Chinese value set to enhance the accuracy and applicability of research outcomes.

CRediT authorship contribution statement

Xu Jin: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Jiaxuan Shi:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Yujia Hu:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Haofei Li:** Writing – review & editing, Project administration, Investigation. **Runhong Li:** Writing – review & editing, Project administration, Investigation. **Yiyin Cao:** Writing – review & editing, Project administration, Investigation. **Zitong Li:** Writing – review & editing, Project administration, Investigation. **Weidong Huang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Xiuzhi Cai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the National Natural Science Foundation of China (No.72274045) and the China National Health Development Research Center Project (CNHRC-KJ-W-2023-27).

References

- [1] Organization WH. Diagnosis and Management of Type 2 Diabetes. WHO. 2020.
- [2] Organization WH. HEARTS D: diagnosis and management of type 2 diabetes. WHO. 2020.
- [3] Federation ID. IDF Diabetes Atlas 10th edition. IDF. 2020.
- [4] Hamman RF, Bell RA, Dabelea D, D'Agostino Jr RB, Dolan L, Imperatore G, et al. The SEARCH for Diabetes in Youth study: rationale, findings, and future directions. *Diabetes Care*. 2014;37:3336–44. <https://doi.org/10.2337/dc14-0574>.

- [5] Duan X, Li Y, Liu Q, Liu L. Epidemiological characteristics, medical costs and healthcare resource utilization of diabetes-related complications among Chinese patients with type 2 diabetes mellitus. *Expert Rev Pharmacoecon Outcomes Res*. 2020;20:513–21. <https://doi.org/10.1080/14737167.2019.1661777>.
- [6] Xiang HD. Study on prevalence rates of hypertension and chronic diabetic complications of in-patients with diabetes. *Chin J Epidemiol* 2003;24:819–21.
- [7] Liu X, Zhang L, Chen W. Trends in economic burden of type 2 diabetes in China: based on longitudinal claim data. *Front Public Health*. 2023;11:1062903. <https://doi.org/10.3389/fpubh.2023.1062903>.
- [8] Mao W, Yip CW, Chen W. Complications of diabetes in China: health system and economic implications. *BMC Public Health*. 2019;19:269. <https://doi.org/10.1186/s12889-019-6569-8>.
- [9] Robinson R. Cost-utility analysis. *Econ Eval Health Care* 1993;307:859–62.
- [10] Brazier J, Ratcliffe J, Saloman J, Tsuchiya A. Measuring and Valuing Health Benefits for Economic Evaluation: Oxford University Press. 2016. doi: 10.1093/med/9780198725923.001.0001.
- [11] Torrance GW. Utility approach to measuring health-related quality of life. *J Chronic Dis*. 1987;40:593–603. [https://doi.org/10.1016/0021-9681\(87\)90019-1](https://doi.org/10.1016/0021-9681(87)90019-1).
- [12] Wang Y, Xu Y, Shan H, Pan H, Chen J, Yang J. Health state utility values of type 2 diabetes mellitus and related complications: a systematic review and meta-regression. *Health Qual Life Outcomes* 2024;22:74. <https://doi.org/10.1186/s12955-024-02288-1>.
- [13] Mok CH, Kwok HHY, Ng CS, Leung GM, Quan J. Health state utility values for Type 2 diabetes and related complications in east and southeast Asia: A systematic review and meta-analysis. *Value Health* 2021;24:1059–67. <https://doi.org/10.1016/j.jval.2020.12.019>.
- [14] Redenz G, Ibaceta MC, Aceituno D, Balmaceda C, Espinoza MA. Health state utility values of type 2 diabetes mellitus and related complications: a systematic review and meta-analysis. *Value Health Reg Issues* 2023;34:14–22. <https://doi.org/10.1016/j.vhri.2022.09.005>.
- [15] Verma S, Leiter LA, Mangla KK, Nielsen NF, Hansen Y, Bonaca MP. Epidemiology and burden of peripheral artery disease in people with type 2 diabetes: a systematic literature review. *Diabetes Ther* 2024;15:1893–961. <https://doi.org/10.1007/s13300-024-01606-6>.
- [16] Hao S, Heintz E, Helgesson G, Langenskiöld S, Chen J, Burstrom K. Influence of elicitation procedure and phrasing on health state valuations in experience-based time trade-off tasks among diabetes patients in China. *Qual Life Res* 2020;29:289–301. <https://doi.org/10.1007/s11136-019-02292-9>.
- [17] Jin X, Liu GG, Gerstein HC, Levine MAH, Guan H, Li H, et al. Minimally important difference and predictors of change in quality of life in type 2 diabetes: a community-based survey in China. *Diabetes Metab Res Rev* 2018;34:e3053. doi: 10.1002/dmrr.3053.
- [18] Gu S, Wang X, Shi L, Sun Q, Hu X, Gu Y, et al. Health-related quality of life of type 2 diabetes patients hospitalized for a diabetes-related complication. *Qual Life Res* 2020;29:2695–704. doi: 10.1007/s11136-020-02524-3.
- [19] Shi Z, Dong YR, Li SP. Health state utility values for diabetes patients in China: a systematic review and meta-analysis. *Modern Preventive Med* 2022;49:1091–8.
- [20] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Bmj*. 2021;372:n71. doi: 10.1136/bmj.n71.
- [21] Muchadeyi MT, Hernandez-Villafuerte K, Schlender M. Quality appraisal for systematic literature reviews of health state utility values: a descriptive analysis. *BMC Med Res Methodol* 2022;22:303. doi: 10.1186/s12874-022-01784-6.
- [22] Nerich V, Saing S, Gamper EM, Holzner B, Pivot X, Viney R, et al. Critical appraisal of health-state utility values used in breast cancer-related cost-utility analyses. *Breast Cancer Res Treat* 2017;164:527–36. doi: 10.1007/s10549-017-4283-8.
- [23] Zoratti MJ, Pickard AS, Stalmeier PFM, Ollendorf D, Lloyd A, Chan KKW, et al. Evaluating the conduct and application of health utility studies: a review of critical appraisal tools and reporting checklists. *Eur J Health Econ*. 2021;22:723–33. doi: 10.1007/s10198-021-01286-0.
- [24] Wei XM, Zhang YG, Tao H, Tang HM, Chen H, Zeng XT. How to perform meta-analysis of correlation coefficients using meta package and metafor package in R. *Chin J Evid-based Med*. 2015;15:850–60. doi: 10.7507/1672-2531.20150144.
- [25] Chen ZC, Zhang SL, Yan L, Wu MC, Chen LH, Ji LN. Association between side effects of oral anti-diabetic drugs and self-reported mental health and quality of life among patients with type 2 diabetes. *Natl Med J China* 2011;91:229–33.
- [26] Yang W, Zhuang X, Li Y, Wang Q, Bian R, Shen J, et al. Improvements in quality of life associated with biphasic insulin aspart 30 in type 2 diabetes patients in China: results from the A1chieve® observational study. *Health Qual Life Outcomes* 2014;12:137. doi: 10.1186/s12955-014-0137-9.
- [27] Ji L, Zou D, Liu L, Qian L, Kadziola Z, Babineaux S, et al. Increasing body mass index identifies Chinese patients with type 2 diabetes mellitus at risk of poor outcomes. *J Diabetes Complications*. 2015;29:488–96. doi: 10.1016/j.jdiacomp.2015.02.014.
- [28] Zhang Y, Wu J, Chen Y, Shi L. EQ-5D-3L Decrements by Diabetes Complications and Comorbidities in China. *Diabetes Ther*. 2020;11:939–50. doi: 10.1007/s13300-020-00788-z.
- [29] Li X, Xu ZR, Wang AH, Niu WF, Li LJ, Mu YM, et al. Quality of Life and Affecting Factors in Elderly Patients with Type 2 Diabetes in Chinese Tertiary Hospitals. *Chin J Mult Organ Dis Elderly*. 2020;19:845–50. doi: 10.11915/j.issn.1671-5403.2020.11.195.
- [30] Li X, Xu ZR, Niu WF, Li LJ, Mu YM, Lu ZQ. Evaluation of the relationship between treatment regimens and quality of life in patients with type 2 diabetes. *Chin J Diabetes*. 2020;28:29–32. doi: 10.1010.3969/j.
- [31] Li C, Zhou H, Wang P. Health utility of type 2 diabetes patients using basal insulin in China: results from the BEYOND II study. *Acta Diabetologica*. 2021;58:329–39. doi: 10.1007/s00592-020-01618-1.
- [32] Xu L. Health-related Quality of Life among Type 2 Diabetics With and Without Complications - A Comparative Research Using EQ-5D Utility Index [Master's thesis]. Jinan: Shandong University; 2017.
- [33] Xu QQ, Guo XL, Chu J, Jing ZY, Zhang XY, Zhou CC. Health-related Quality of Life and Its Influencing Factors in Type 2 Diabetes Patients in Shandong Province. *Journal of shandong university(health sciences)*. 2019;57:96–103 doi: 10.6040/j.issn.1671-7554.0.2018.1130.
- [34] Gu S, Wang X, Shi L, Sun Q, Hu X, Gu Y, et al. Health-related quality of life of type 2 diabetes patients hospitalized for a diabetes-related complication. *Qual Life Res*. 2020;29:2695–704. doi: 10.1007/s11136-020-02524-3.
- [35] Liu Y, Lin YT, Qi YT, Gao J, Geng XE, Cui Y, et al. Health-related Quality of Life in Patients with Type 2 Diabetes Based on the European 5-Dimensional Health Scale. *J Pub Health Prev Med*. 2023;34:11–5. doi: 10.3969/j.issn.1006-2483.2023.04.003.
- [36] Du HM. Moderating the Relationship of Self-empowerment and Quality-of-life among Type 2 Diabetic Patients in Shandong: the Role of Education Level [Master's degree]. Shandong: Shandong University; 2023. doi: 10.27272/d.cnki.gshdu.2023.004238.
- [37] Li MT, He Y. Medical coping style and its effects on quality of life of type 2 diabetes patients in Wuxi. *Med Soc* 2019;32. 89–92+130.
- [38] Wang S. Prevalence, Risk Factors and Effects on General Quality of Life in Patients with Type 2 Diabetes Mellitus with Diabetic Retinopathy in an Urban Community [Master's degree]. Suzhou: Suzhou University; 2019. doi: 10.27351/d.cnki.gszhu.2019.001142.
- [39] Wang J, Li JJ, Zhang Y, Fan K, Zhang W, Zhang H, et al. Assessment of quality of life and its related factors in patients with type 2 diabetes mellitus. *Chin J Gerontol* 2018;38:4292–5.
- [40] Wang DW. Effect of measures to improve doctor's quality on blood sugar management in patients with type 2 diabetes mellitus [Master's degree]. Hefei: Anhui Medical University; 2019.
- [41] Kang N, Liu X, Liao W, Tu R, Sang S, Zhai Z, et al. Health-related quality of life among rural adults with type 2 diabetes mellitus: a cross-sectional study. *Eur J Public Health* 2021;31:547–53. doi: 10.1093/eurpub/ckaa247.
- [42] Liu QL. Prevalence and Influencing Factors of Fear of Hypoglycemia in Hospitalized patients with Type 2 Diabetes Mellitus [Master's degree]. Urumqi: Xinjiang Medical University; 2022. doi: 10.27433/d.cnki.gxyku.2022.000454.
- [43] Lin P, Liu Z, Hong X, Chen X, Liu J. Evaluation and analysis of the quality of life of type 2 diabetic patients, Hainan. *Modern Preventive Med* 2023;50:1509–15. doi: 10.20043/j.cnki.MPM.202207170.
- [44] Lu Y, Wang N, Chen Y, Nie X, Li Q, Han B, et al. Health-related quality of life in type-2 diabetes patients: a cross-sectional study in East China. *BMC Endocr Disord* 2017;17:38. doi: 10.1186/s12902-017-0187-1.
- [45] Pan CW, Wang S, Wang P, Xu CL, Song E. Diabetic retinopathy and health-related quality of life among Chinese with known type 2 diabetes mellitus. *Qual Life Res* 2018;27:2087–93. doi: 10.1007/s11136-018-1876-6.
- [46] Pan CW, Sun HP, Zhou HJ, Ma Q, Xu Y, Luo N, et al. Valuing health-related quality of life in type 2 diabetes patients in China. *Med Decis Making* 2016;36:234–41. doi: 10.1177/0272989x15606903.
- [47] Zhuang Y, Ma QH, Pan CW, Lu J. Health-related quality of life in older Chinese patients with diabetes. *PLoS One* 2020;15:e0229652. doi: 10.1371/journal.pone.0229652.
- [48] Ying L, Guan H, Liu G, Sun L. The influences of alcohol consumption on the life quality of patients with type 2 diabetes mellitus in China. *Chinese Health Econ* 2017;36:74–6. doi: 10.7664/che20170621.
- [49] Han Y, Wu J, Cai D, Yin S, Zhuang X, Li C, et al. Evaluation of health-related quality of life in type 2 diabetic patients with various complications and oral drug treatment China. *J Endocrinol Metab* 2013;29. <https://doi.org/10.3760/cma.j.issn.1000-6699.2013.08.005>.
- [50] Tang Z, Jiang X, Hong L, Feng Z, He Q, Yuan J, et al. Validation of the simplified chinese version of the brief diabetes quality of life (DQoL) questionnaire based on a cross-sectional study. *Int J Environ Res Public Health* 2020;17. doi: 10.3390/ijerph17238792.
- [51] Zhang Y, Sun J, Pang Z, Wang X, Gao W, Ning F, et al. The impact of new screen-detected and previously known type 2 diabetes on health-related quality of life: a population-based study in Qingdao, China. *Qual Life Res*. 2014;23:2319–26. doi: 10.1007/s11136-014-0674-z.
- [52] Huang SG. Research on the Relationships among Fear of Hypoglycemia, Self-efficacy and Quality of Life in Patients with Type 2 Diabetes Mellitus [Master's degree]. Changchun: Jilin University; 2017.
- [53] Hao S, Heintz E, Helgesson G, Langenskiöld S, Chen J, Burström K. Influence of elicitation procedure and phrasing on health state valuations in experience-based time trade-off tasks among diabetes patients in China. *Qual Life Res* 2020;29:289–301. doi: 10.1007/s11136-019-02292-9.
- [54] Carter N, Li J, Xu M, Li L, Fan X, Zhu S, et al. Health-related quality of life of people with type 2 diabetes and its associated factors at a tertiary care clinic in Ningbo, China: a cross-sectional study. *Endocrinol Diabetes Metab* 2022;5:e353. doi: 10.1002/edm2.353.
- [55] Zhao M, Ren Y, Tang P, Cao W, Huang Y, Gong F. Study on health-related quality of life of patients with type 2 diabetes and its complications in Shenzhen City. *Occup Health* 2023;39(221–6):32.
- [56] Su SH. Study on the current situation and related factors of fear of hypoglycemia among elderly patients with type 2 diabetes mellitus based on the theory of

- unpleasant symptoms [Master's degree]. Chengdu: Chengdu University of TCM; 2023. doi: 10.26988/d.cnki.gcdzu.2023.000615.
- [57] Xie K. Research on quality of life and influencing factors in patients with type 2 diabetes mellitus in China [Master's degree]. Tianjin: Tianjin University; 2014.
- [58] Pan CW, Sun HP, Wang X, Ma Q, Xu Y, Luo N, et al. The EQ-5D-5L index score is more discriminative than the EQ-5D-3L index score in diabetes patients. *Qual Life Res*. 2015;24:1767–74. doi: 10.1007/s11136-014-0902-6.
- [59] Afshari S, Ameri H, Baharinya S, Arab-Zozani M, Mojahedian MM. Assessment of the properties of the EQ-5D-5L in patients with type 2 diabetes mellitus: a systematic review and meta-analysis. *Expert Rev Pharmacoecon Outcomes Res* 2022;22:351–64. doi: 10.1080/14737167.2022.2011216.
- [60] Tao X, Xue Y, Niu R, Lu W, Yao H, He C, et al. Comparisons of clinical outcomes between newly diagnosed early- and late-onset T2DM: a real-world study from the Shanghai Hospital Link Database. *Endocr Connect*. 2024;13. doi: 10.1530/ec-23-0474.
- [61] Hayes A, Arima H, Woodward M, Chalmers J, Poulter N, Hamet P, et al. Changes in quality of life associated with complications of diabetes: results from the ADVANCE study. *Value Health*. 2016;19:36–41. doi: 10.1016/j.jval.2015.10.010.
- [62] Shao H, Yang S, Fonseca V, Stoecker C, Shi L. Estimating quality of life decrements due to diabetes complications in the united states: the health utility index (HUI) diabetes complication equation. *Pharmacoeconomics*. 2019;37:921–9. doi: 10.1007/s40273-019-00775-8.
- [63] Kuo S, Yang C-T, Chen H-Y, Ou H-T. Valuing health states of people with type 2 diabetes: analyses of the nationwide representative linked databases. *J Diab Invest* 2021;12:1749–58. doi: 10.1111/jdi.13520.
- [64] Lung TW, Hayes AJ, Hayen A, Farmer A, Clarke PM. A meta-analysis of health state valuations for people with diabetes: explaining the variation across methods and implications for economic evaluation. *Qual Life Res* 2011;20:1669–78. doi: 10.1007/s11136-011-9902-y.
- [65] Committee ADAPP. Older adults: standards of medical care in diabetes. *Diabetes Care*. 2022;45:S195–S207. doi: 10.2337/dc22-S013.
- [66] Takahara M, Katakami N, Shiraiwa T, Abe K, Ayame H, Ishimaru Y, et al. Evaluation of health utility values for diabetic complications, treatment regimens, glycemic control and other subjective symptoms in diabetic patients using the EQ-5D-5L. *Acta Diabetol* 2019;56:309–19. doi: 10.1007/s00592-018-1244-6.
- [67] Bansback N, Brazier J, Tsuchiya A, Anis A. Using a discrete choice experiment to estimate health state utility values. *J Health Econ* 2012;31:306–18. doi: 10.1016/j.jhealeco.2011.11.004.
- [68] Torrance GW. Measurement of health state utilities for economic appraisal. *J Health Econ* 1986;5:1–30. doi: 10.1016/0167-6296(86)90020-2.
- [69] Nikl A, Janssen MF, Brodzsky V, Rencz F. A head-to-head comparison of the EQ-5D-5L and 15D descriptive systems and index values in a general population sample. *Health Qual Life Outcomes* 2023;21:17. doi: 10.1186/s12955-023-02096-z.
- [70] Sayah FA, Qiu W, Xie F, Johnson JA. Comparative performance of the EQ-5D-5L and SF-6D index scores in adults with type 2 diabetes. *Qual Life Res* 2017;26:2057–66. doi: 10.1007/s11136-017-1559-8.
- [71] Gundle KR, Cizik AM, Punt SE, Conrad EU, 3rd, Davidson DJ. validation of the SF-6D health state utilities measure in lower extremity sarcoma. *Sarcoma*. 2014;2014:450902. doi: 10.1155/2014/450902.
- [72] Ye Z, Sun L, Wang Q. A head-to-head comparison of EQ-5D-5 L and SF-6D in Chinese patients with low back pain. *Health Qual Life Outcomes* 2019;17:57. doi: 10.1186/s12955-019-1137-6.
- [73] Xie S, Wang D, Wu J, Liu C, Jiang W. Comparison of the measurement properties of SF-6Dv2 and EQ-5D-5L in a Chinese population health survey. *Health Qual Life Outcomes* 2022;20:96. doi: 10.1186/s12955-022-02003-y.
- [74] Herdman M, Gudex C, Lloyd A, Janssen M, Kind P, Parkin D, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). *Qual Life Res* 2011;20:1727–36. doi: 10.1007/s11136-011-9903-x.
- [75] Janssen MF, Pickard AS, Golicki D, Gudex C, Niewada M, Scalone L, et al. Measurement properties of the EQ-5D-5L compared to the EQ-5D-3L across eight patient groups: a multi-country study. *Qual Life Res* 2013;22:1717–27. doi: 10.1007/s11136-012-0322-4.
- [76] Jiang J, Hong Y, Zhang T, Yang Z, Lin T, Liang Z, et al. Comparing the measurement properties of the EQ-5D-5L and the EQ-5D-3L in hypertensive patients living in rural China. *Qual Life Res* 2021;30:2045–60. doi: 10.1007/s11136-021-02786-5.
- [77] Weng G, Hong Y, Luo N, Mukuria C, Jiang J, Yang Z, et al. Comparing EQ-5D-3L and EQ-5D-5L in measuring the HRQoL burden of 4 health conditions in China. *Eur J Health Econ* 2023;24:197–207. doi: 10.1007/s10198-022-01465-7.
- [78] Tai Z, Wan D, Zan Q, Huang Y, Xu C. Comparison of the EQ-5D-3L and EQ-5D-5L instruments in patients undergoing unicompartmental knee arthroplasty. *Front Med (Lausanne)*. 2024;11:1451979. doi: 10.3389/fmed.2024.1451979.
- [79] Zhao Y, Li SP, Liu L, Zhang JL, Chen G. Does the choice of tariff matter?: a comparison of EQ-5D-5L utility scores using Chinese, UK, and Japanese tariffs on patients with psoriasis vulgaris in Central South China. *Medicine (Baltimore)*. 2017;96:e7840. doi: 10.1097/MD.00000000000007840.
- [80] Sullivan PW, Ghushchyan VH. EQ-5D scores for diabetes-related comorbidities. *Value Health*. 2016;19:1002–8. <https://doi.org/10.1016/j.jval.2016.05.018>.